ML COURSE

NTI

Intelligent Multi-Condition Medical Prediction System

PROJECT OVERVIEW

Our system handles three different medical prediction tasks:

Medical Condition	Data Type	Prediction Task	Output
🫀 ECG	ECG Signals	Multiclass Classification	Heartbeat Class
❤️ Heart Failure	Clinical Data	Binary Classification	Death / No Death
🧪 CKD	Clinical Data	Binary Classification	CKD / No CKD

ONE AI SYSTEM → THREE MEDICAL PREDICTION TASKS

Problem Statement

Medical data can be:

- Large and complex
- Difficult to analyze manually
- Time-consuming to interpret
- Rich in hidden patterns

Our Solution

Use AI to learn patterns from medical data and provide predictions.

PROJECT OBJECTIVES

- Analyze three medical datasets
- Perform EDA and data cleaning
- Apply appropriate preprocessing techniques
- Train multiple ML and DL models

- Evaluate and compare model performance
- Select the best model for each task
- Deploy the models into one web application

DATASETS

Dataset	Data Type	Task	Target
ECG	Signal	Multiclass Classification	5 Classes
Heart Failure	Tabular	Binary Classification	DEATH_EVENT
CKD	Tabular	Binary Classification	CKD / No CKD

- **Signal = data representing a changing signal over time**
- **Tabular = data organized into rows and columns**

Project Workflow:

MEDICAL DATA INPUT (Telemetry & Tabular Records)

↓

ECG Pipeline
- Preprocessing (Scaling & R-Peak Filtering)
- Model: Gradient Boosting / RF
- Evaluation & Waveform Plotting

Heart Failure
- Preprocessing (Selective Scaling)
- Model: Survival Classifier
- Evaluation & Risk Gauge

CKD Diagnostic
- Preprocessing (Imputation & Encoding)
- Model: 100% Recall Classifier
- Evaluation & Nephrology Verdict

↓

UNIFIED WEB DEPLOYMENT → CLINICAL PREDICTION OUTPUT

OUR AI APPROACH

Supervised Learning Architecture

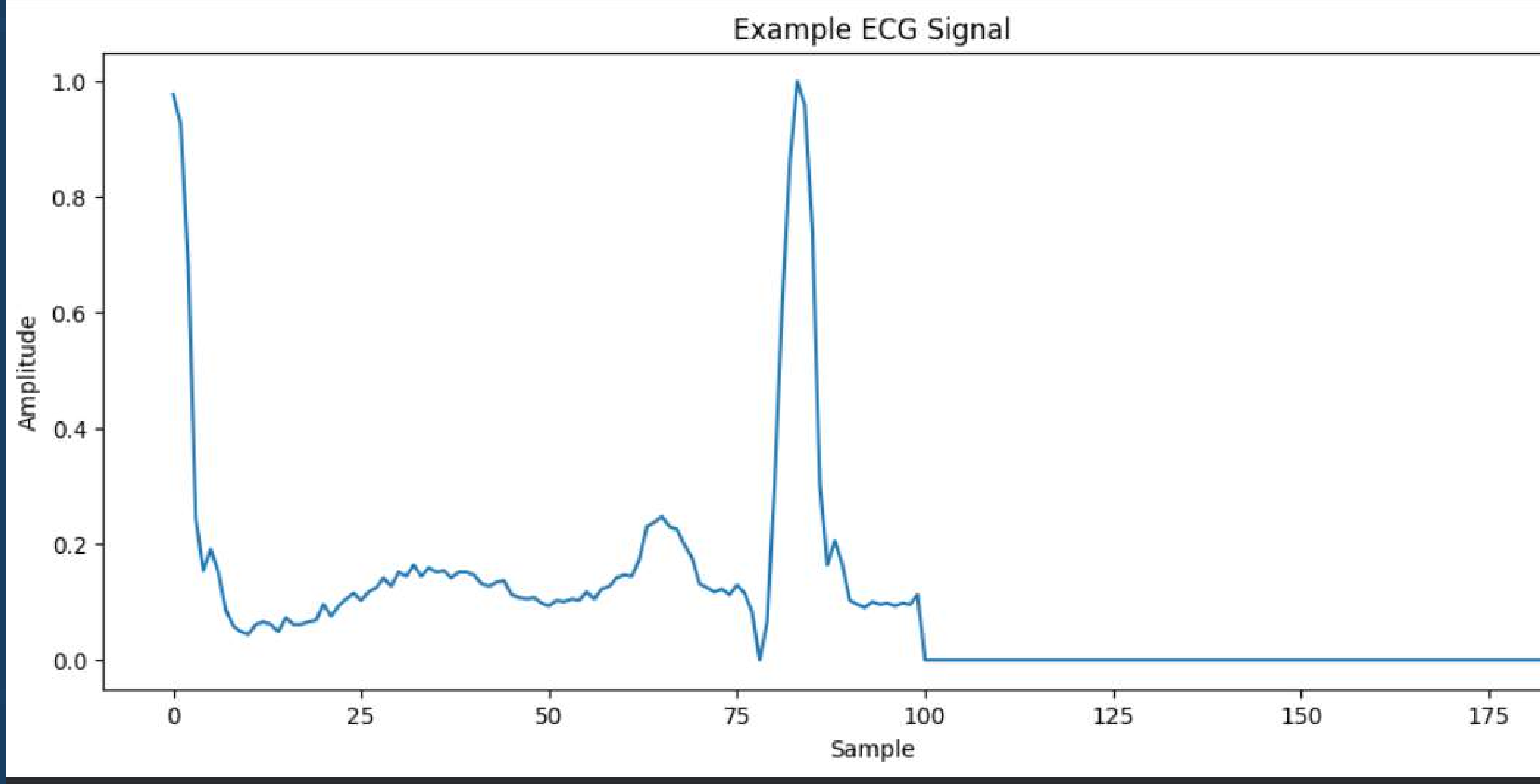
- Algorithmic Paradigm: Supervised learning framework mapping historical clinical parameters directly to validated diagnostic outputs.
- Task Formulation: Formulated as robust classification pipelines predicting discrete medical endpoints and pathological classes.
- Inference Objective: Optimizing high-dimensional decision boundaries to accurately categorize complex physiological signals and patient biometrics.

Clinical Target Taxonomy

- ECG Arrhythmia Telemetry: Multiclass classification mapping 188-sample time-series waveforms across 5 distinct cardiac beat categories.
- Heart Failure Survival Risk: Binary classification evaluating patient survival likelihood versus adverse mortality outcomes.
- Chronic Kidney Disease (CKD): Binary classification distinguishing healthy renal function from pathological CKD presentation across 24 clinical biomarkers.

ECG Heartbeat Classification

Dataset: MIT-BIH Arrhythmia Dataset
Data Type: ECG Signal Data
The dataset contains 5 heartbeat classes
Task: Multiclass Classification



Output:

5 heartbeat classes
Class 0
Class 1
Class 2
Class 3
Class 4

Input:

- ECG heartbeat signal represented by numerical values

ECG – EDA

- **Class Distribution Analysis**
- **Data Visualization**
- **Class Imbalance Detection**

Class 0 → 72,024 (highest class)

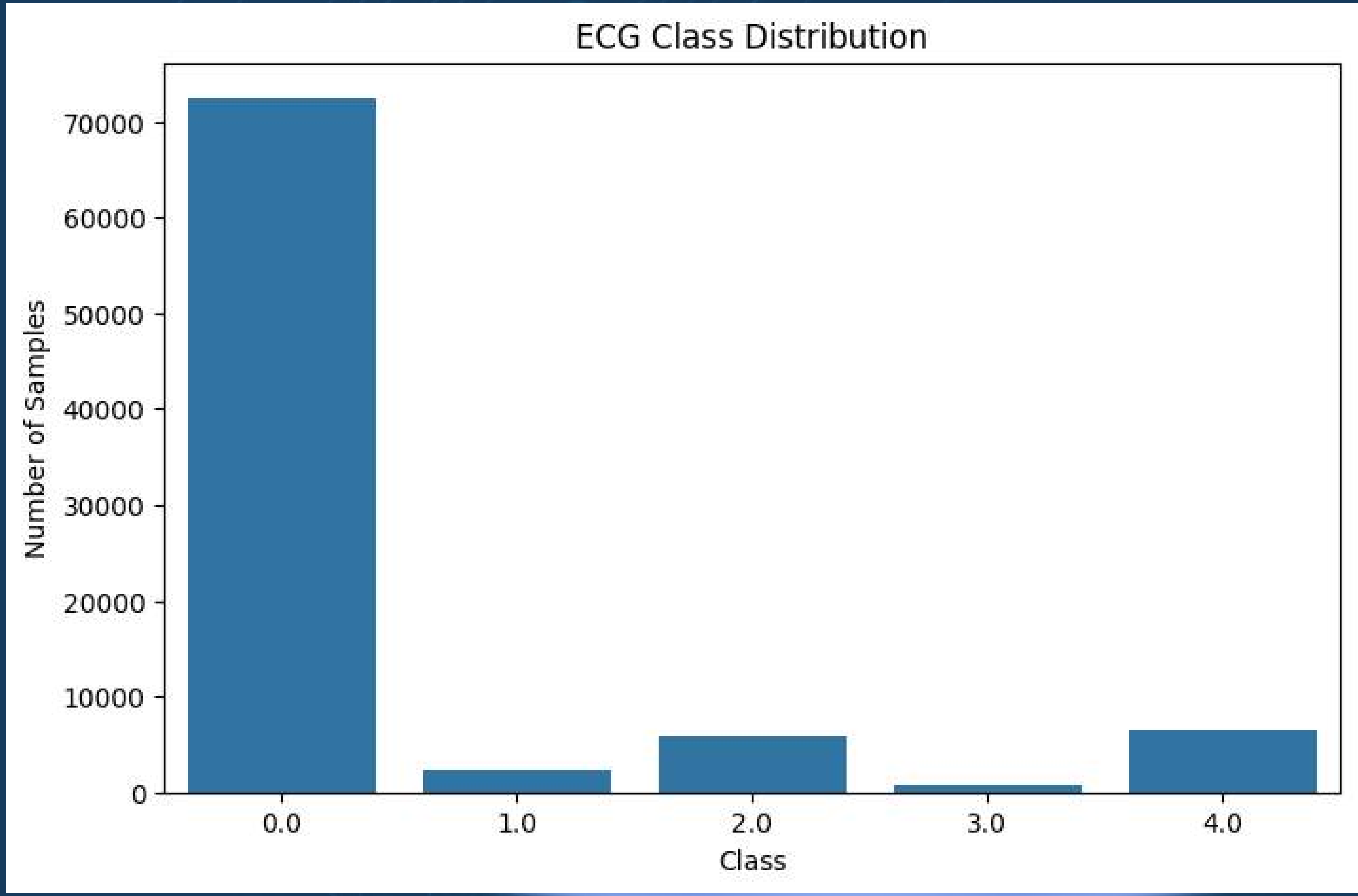
Class 1 → 2,223

Class 2 → 5,788

Class 3 → 641 (lowest class)

Class 4 → 6,431

The ECG dataset is highly imbalanced



ECG Model

ECG Preprocessing:

Feature & Target Separation:

- X → ECG signal features
- y → Heartbeat class

Train-Test Split:

- 80% Training
- 20% Testing

ECG Data Cleaning:

- Missing values: 0
- Infinite values: 0
- Duplicates: Removed
- Invalid Data

Machine Learning Models

We evaluated:

- Logistic Regression
- K-Nearest Neighbors
- Decision Tree
- Random Forest
- SVM (SVC for Classification)

Model	Main Idea
Logistic Regression	Baseline linear classifier use probability
KNN	Uses nearby samples
Decision Tree	Rule-based decisions
Random Forest	Multiple decision trees
SVM	Finds separating boundaries

	Model	Accuracy	Precision	Recall	F1-Score
0	Logistic Regression	0.673990	0.881088	0.673990	0.737215
1	KNN	0.972730	0.971766	0.972730	0.971559
2	Decision Tree	0.955235	0.954890	0.955235	0.955053
3	Random Forest	0.973004	0.972855	0.973004	0.971147
4	SVM	0.879088	0.897639	0.879088	0.886884

Best Performing Model: Random Forest

Deep Learning

ECG — Deep Learning Models:

- **Neural Network** (**NN**)
  Learns complex relationships between input features
- **Convolutional Neural Network** (**CNN**)
  Extracts important patterns from ECG signals
- **Recurrent Neural Network** (**RNN**)
  Processes sequential ECG data
- **Long Short-Term Memory** (**LSTM**)
  Captures long-term dependencies in ECG sequences

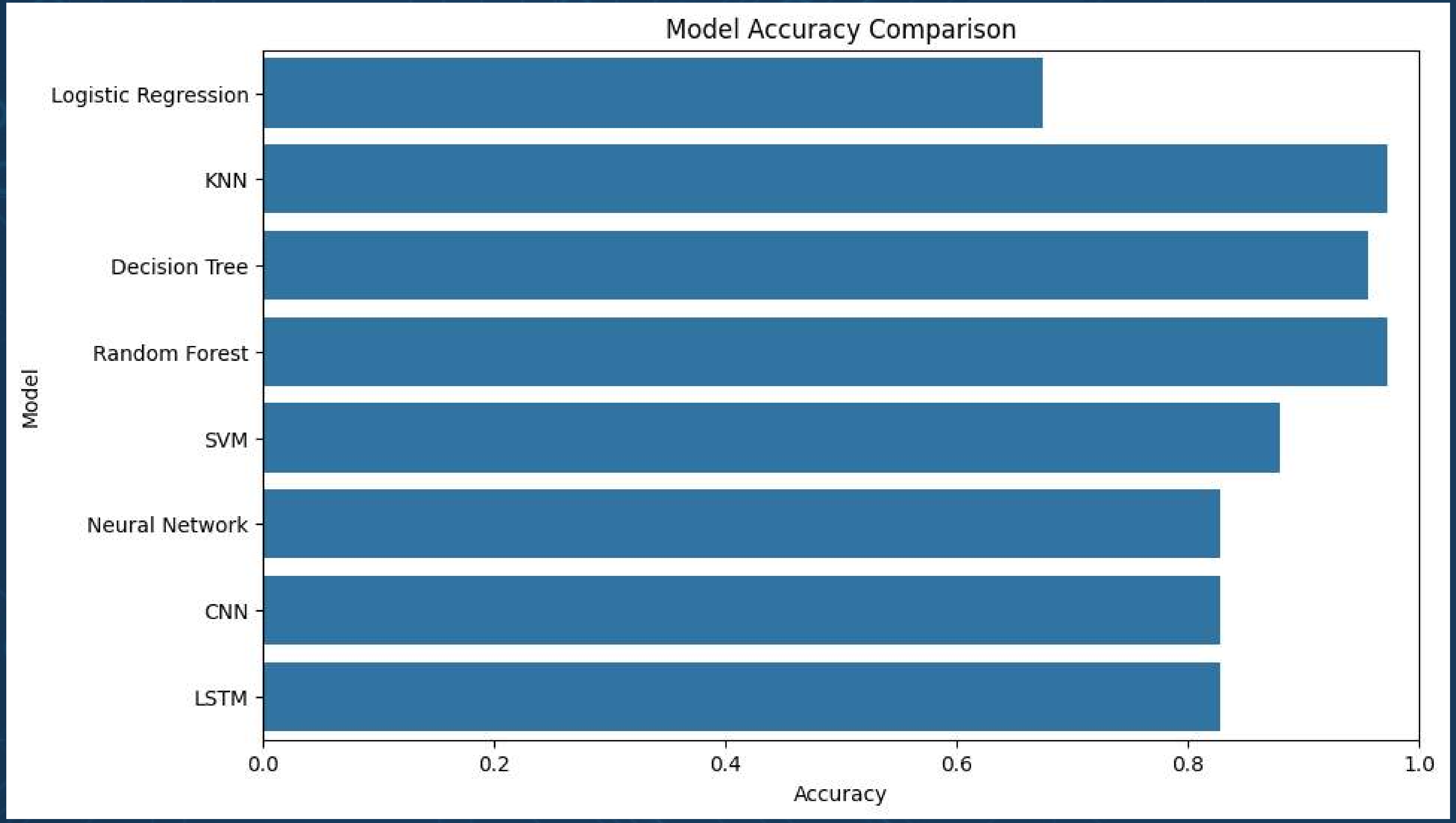
ECG Results

Model Evaluation:

- Recall
- F1-Score
- Confusion Matrix
- Accuracy
- Precision

Best Model:

- Random Forest
  - Accuracy: 97.30%
  - Precision: 97.29%
  - Recall: 97.30%
  - F1-Score: 97.11%



Heart Failure Prediction

Dataset Source & Structure

- Features: 12 Clinical Variables
- Target Variable: DEATH_EVENT

0 = Survived , 1 = Died

Heart Function & Enzymes

- Ejection Fraction
- CPK Enzyme

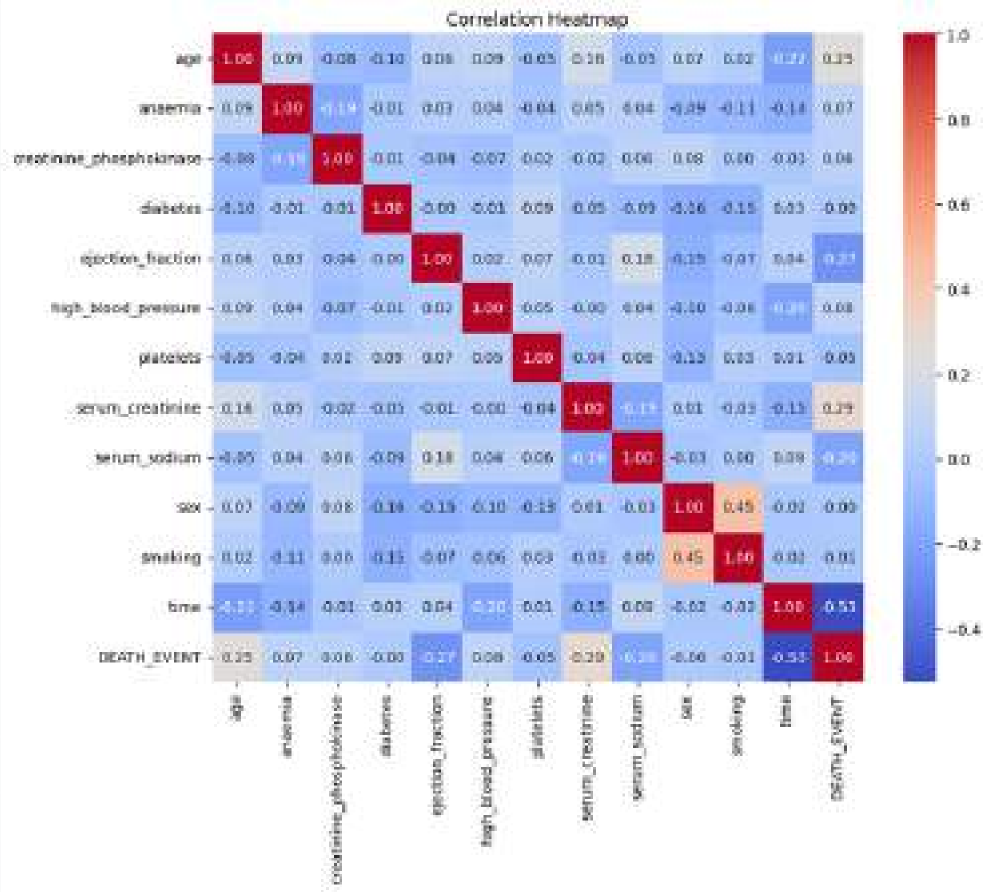
Systemic & Renal Biomarkers

- Serum Creatinine
- Serum Sodium
- Blood Pressure
- Platelets

Risk Factors

- Diabetes
- Anemia
- Smoking
- Age Gender
- Follow-up time

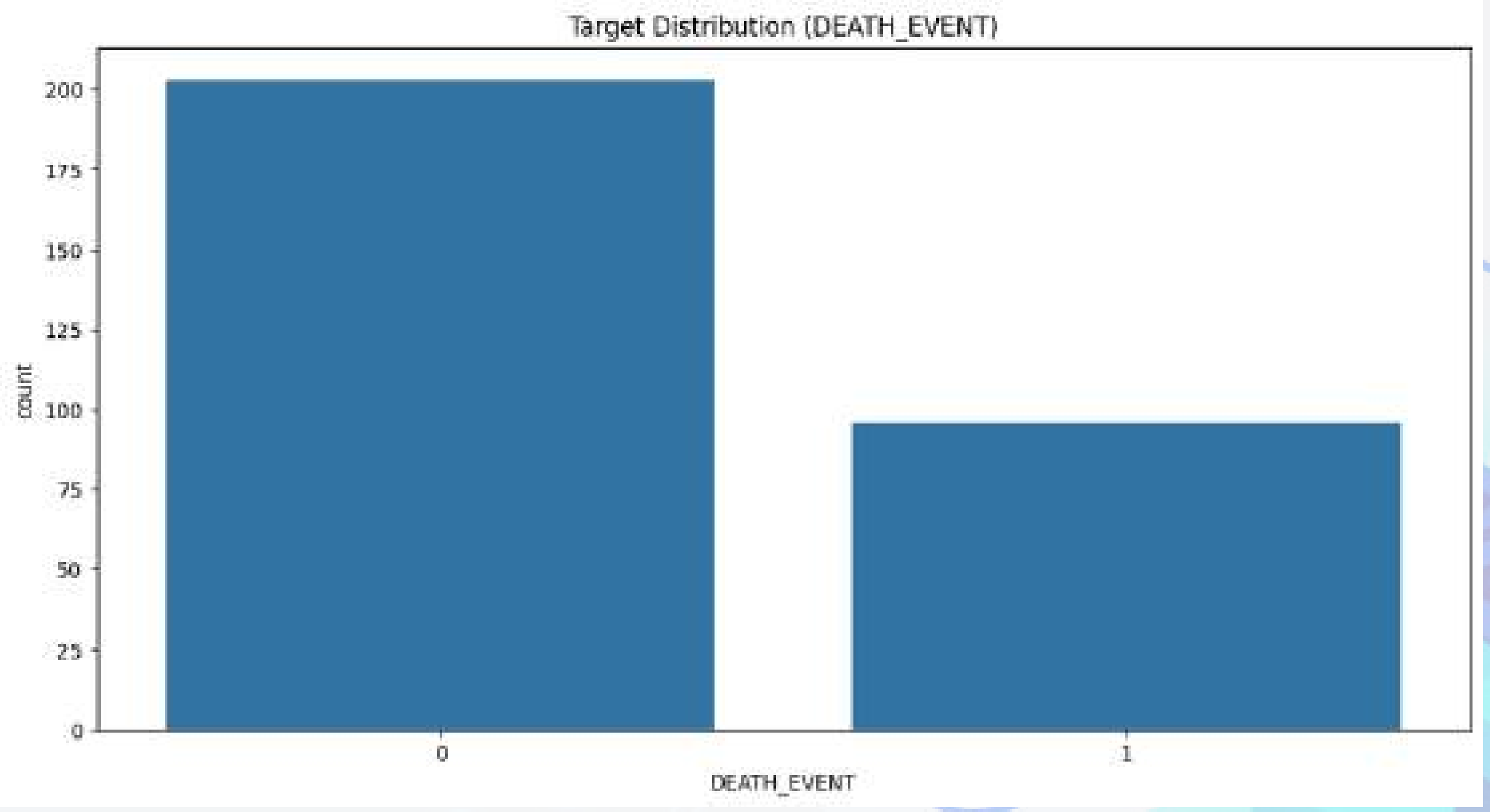
Heart EDA



Key Feature Insights:

Time & Ejection Fraction: Strongly linked to higher survival (increases chance of survival)

Serum Creatinine & Age: Highest direct risk factors (increases mortality rate)



Target Distribution

(DEATH_EVENT)

0 (Survived): ~68% (203 patients) — Majority class.

1 (Died): ~32% (96 patients) — Imbalanced target.

Heart Failure Preprocessing

- Target Variable: DEATH_EVENT
- Outliers Check: Verified extreme values (Kept as valid medical indicators).
- Feature Scaling: Applied StandardScaler to normalize numeric ranges.
- Data Split: Training Set => 80%
  Testing Set =>20% (Stratified Split)

Cleaning Data

```
df.isnull().sum()
```

	0
age	0
anaemia	0
creatinine_phosphokinase	0

- Missing Values: 0 missing values
- (Dataset is 100% complete)

```
df.duplicated().sum()
```

```
np.int64(0)
```

- Duplicate Records: 0 duplicate rows.

MODELING

=== Machine Learning Models Performance Comparison ===

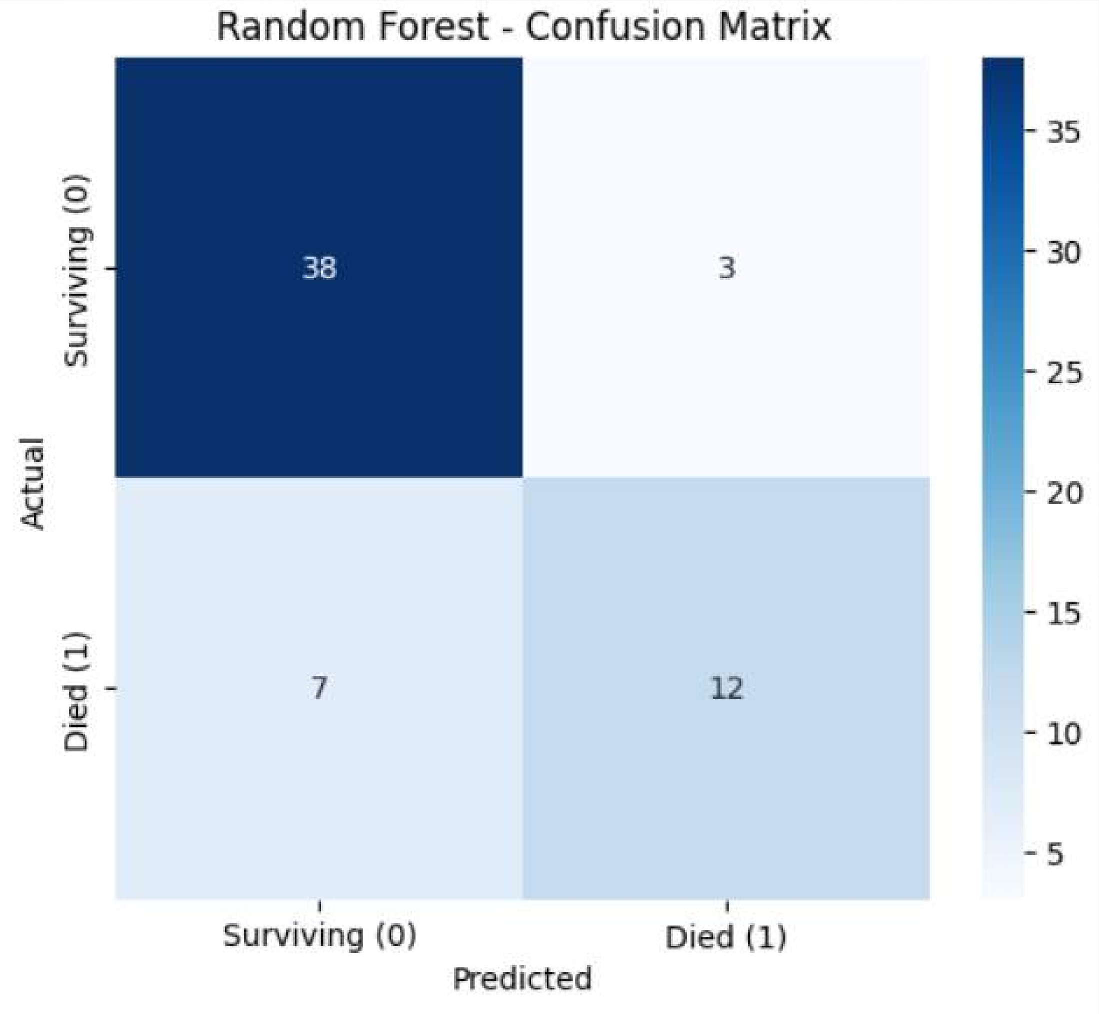
Model	Accuracy (%)
Gradient Boosting	83.33
Random Forest (Best)	83.33
XGBoost	81.66
Logistic Regression	81.66
AdaBoost	80.00
Bagging	78.33
SVM	76.66
Decision Tree	73.33
KNN	71.66
Naive Bayes	70.00

- **Model Exploration: Evaluated 10 classification algorithms**
  (Linear, Tree-based, Ensembles).

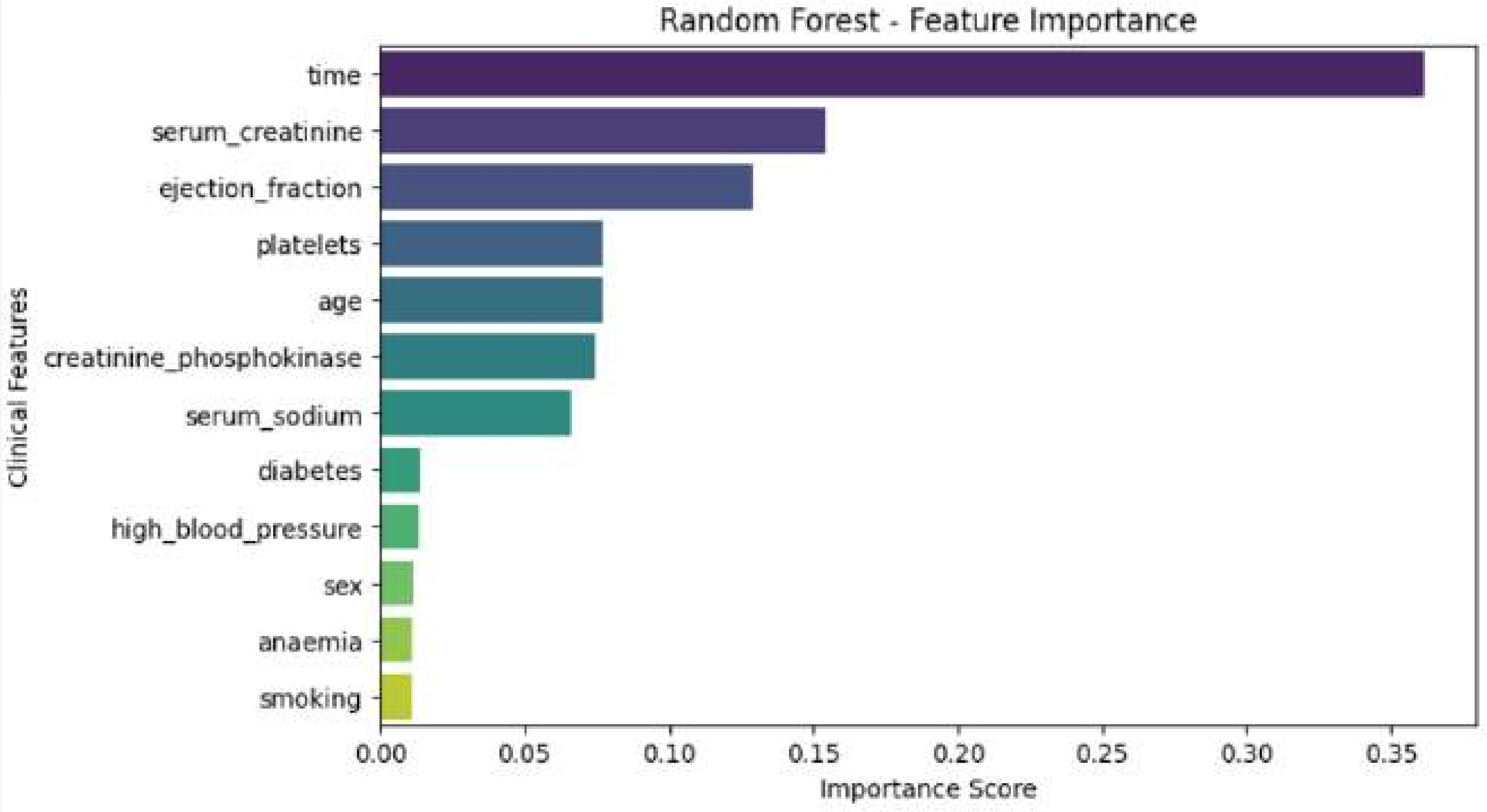
- **Top Performers: Gradient Boosting, Random Forest and XGBoost achieved highest accuracies >=81.66%.**

Model	**Key Focus / Characteristics**
Logistic Regression	Baseline linear model for direct medical interpretability.
KNN & SVM	Distance & margin boundary classifiers (simpler baseline).
Decision Tree	Single rule-based classifier (prone to overfitting).
Random Forest & Boosting	Ensembles using trees to boost overall accuracy & reduce error.

BEST MODEL



- Random Forest Performance (Best Model)
- Highest Accuracy: Achieved 83.33% overall accuracy.
- Confusion Matrix: Correctly predicted 38 surviving and 12 mortality cases.



- Top Feature Importance
- Primary Driver: time is the most influential variable in prediction.
- Key Indicators: serum_creatinine and ejection_fraction are the 2nd & 3rd top features.

Deep Learning

<u>ANN Model Architecture & Training</u>

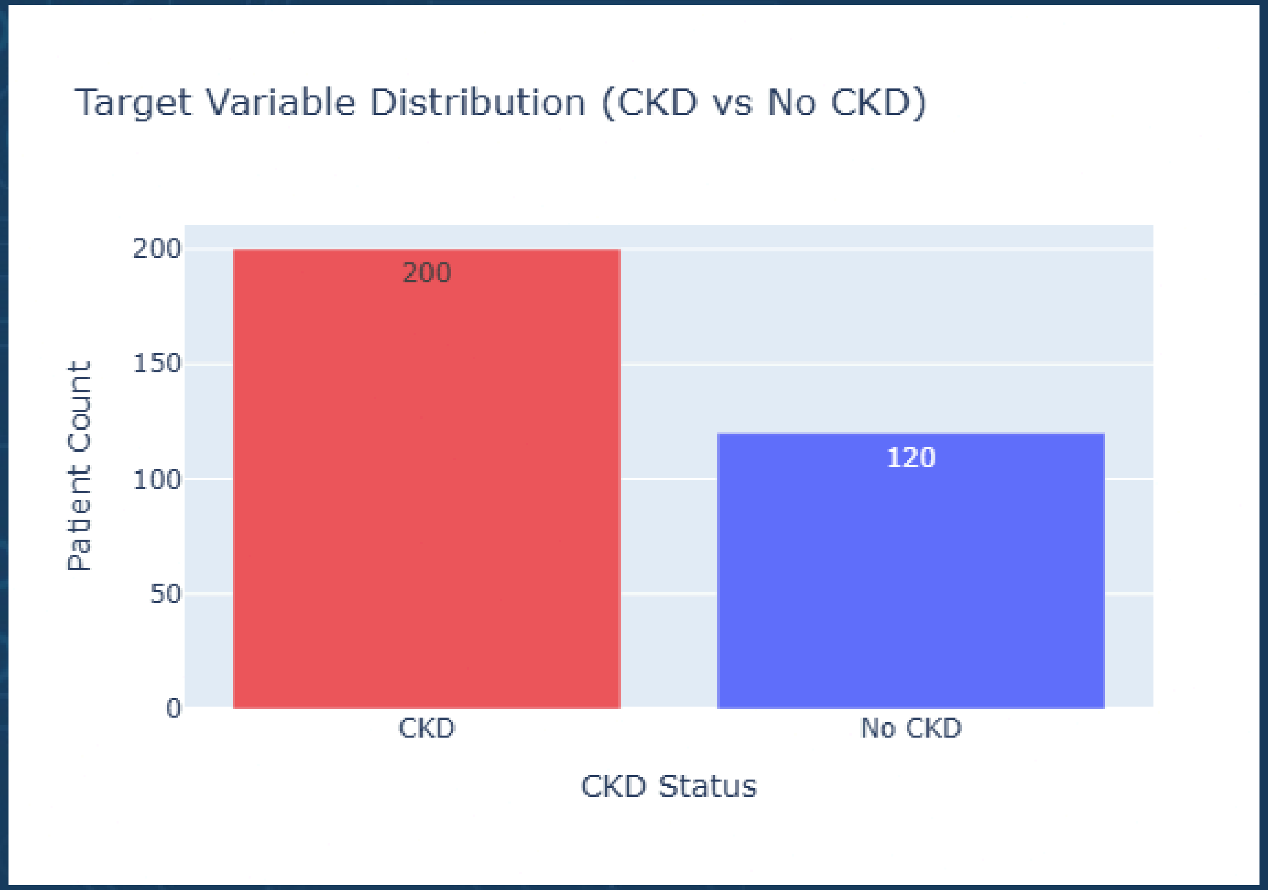
- Network Structure: Built a <u>*Sequential*</u> Artificial Neural Network (ANN) with <u>*Dense*</u> layers.
- Overfitting Control: Integrated <u>*Dropout*</u> layers and <u>*EarlyStopping*</u> to optimize training rounds.
- Optimization: Used <u>*Adam*</u> optimizer with Binary Crossentropy loss function.
- Test Accuracy: Achieved <u>*85% accuracy*</u> on the testing set.

CKD Dataset

- Objective:
  - Early diagnostic classification of Chronic Kidney Disease using multi-modal physiological and biochemical markers.
- Data Hygiene & Schema:
  - Resolved ARFF formatting corruption across 400 patient profiles with 24 clinical features (11 continuous, 13 categorical) and a binary target of 250 CKD and 150 Not-CKD.
- Anti-Leakage Protocol:
  - Executed an 80/20 stratified split (320 train, 80 test) prior to any data transformation, imputation, or statistical analysis.

EDA and Insights

- **Distribution & Class Imbalance**
  - Stratified training baseline reflects 200 CKD (62.5%) vs 120 healthy patients (37.5%), preserving exact population proportions before modeling.
- **Biochemical Skew & Waste Accumulation**
  - Extreme positive skewness identified in primary metabolic waste indicators: Serum Creatinine (sc = 5.03), Blood Urea (bu = 2.87), and Potassium (pot = 10.51).
- **The Clinical Outlier Insight**
  - 100% of IQR outliers across key markers belonged exclusively to CKD patients. Outliers represent biological pathology rather than noise, validating tree-based partitioning.



Model Peformance

Key Peformance:

- Top Tier Performance: Random Forest and AdaBoost tie for the best F1-Score (99.01%), both at 98.75% Accuracy with 100% Recall. No missed CKD cases.
- Strong Ensembles: Random Forest, AdaBoost, and SVM all reach 98.75% Accuracy with Random Forest and AdaBoost tying for the top F1-Score (99.01%).
- Recall Consistency: Multiple models (Random Forest, Bagging, Gradient Boosting, XGBoost, Decision Tree) achieve 100.0% Recall, ensuring minimal false negatives in disease detection.

Model Selection Rationale

- Selection Rule: Since a missed CKD diagnosis is far costlier than a false alarm, we required 100% Recall as a non-negotiable filter first, then used F1-Score as the tiebreaker not accuracy alone, which would be misleading on a 62.5/37.5 class split.
- Reliable Generalization: We chose the tuned Random Forest over the tied untuned Random Forest because 5-fold cross-validation gave it a more robust decision boundary (99.51% CV F1), trading a small amount of test F1 (99.01% → 98.04%) for expected stability on unseen patients.
- Clinical Explainability: Random Forest preserves transparent feature importance rankings and auditability, allowing clinicians to verify core predictive biomarkers like hemoglobin and serum creatinine.

	Accuracy	Precision	Recall	F1-Score
Random Forest	98.75%	98.04%	100.0%	99.01%
AdaBoost	98.75%	98.04%	100.0%	99.01%
SVM	98.75%	100.0%	98.0%	98.99%
Random Forest (Tuned)	97.5%	96.15%	100.0%	98.04%
Bagging	97.5%	96.15%	100.0%	98.04%
Gradient Boosting	97.5%	96.15%	100.0%	98.04%
XGBoost	97.5%	96.15%	100.0%	98.04%
Logistic Regression	97.5%	98.0%	98.0%	98.0%
Naive Bayes	97.5%	100.0%	96.0%	97.96%
Decision Tree	96.25%	94.34%	100.0%	97.09%
KNN	95.0%	97.92%	94.0%	95.92%

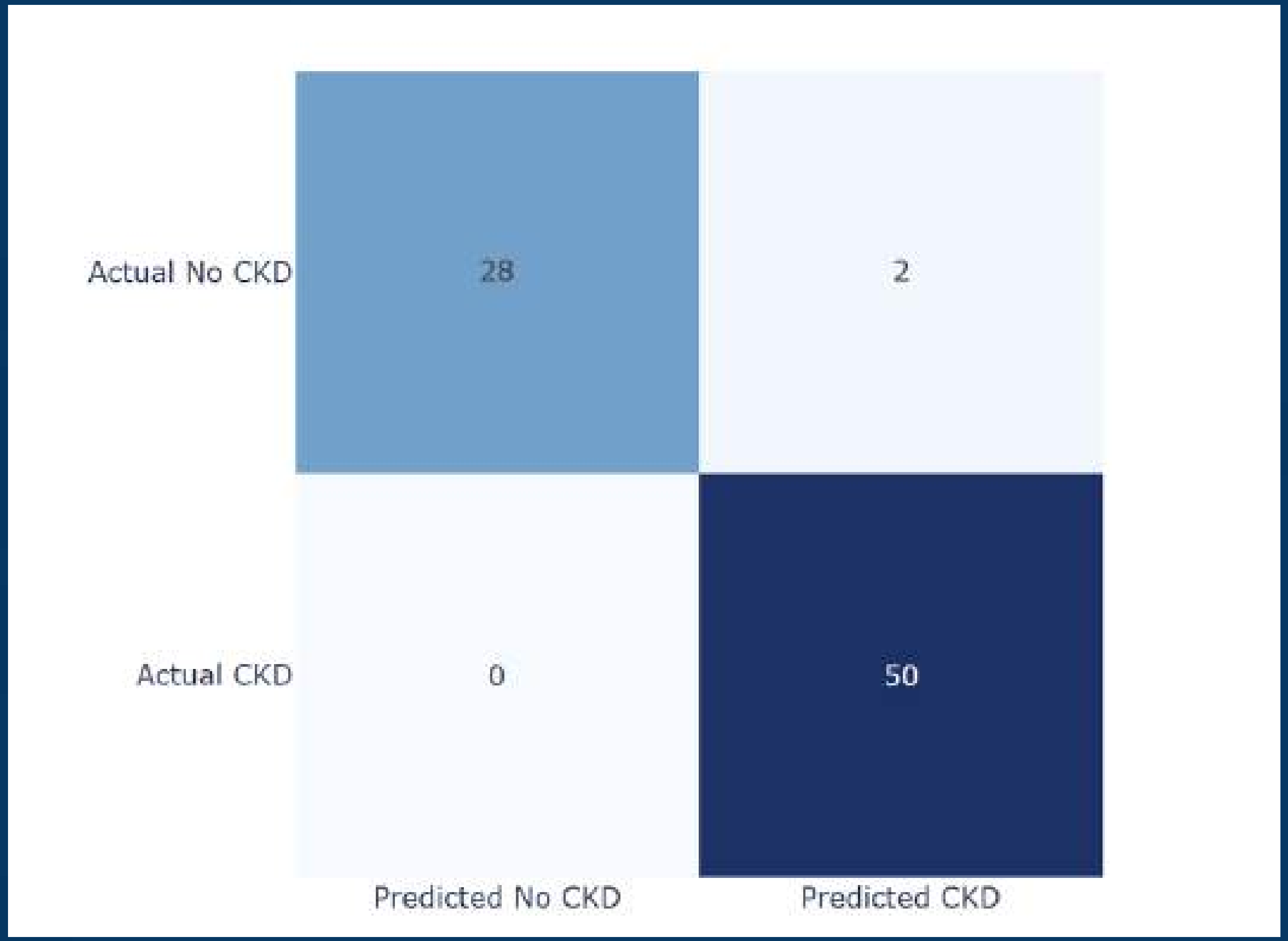
RF Model Evaluation

Holdout Performance (N = 80):

- Overall Test Accuracy: 97.5% (78 / 80 correct classifications).
- True Positives (CKD): 50
- True Negatives (No CKD): 28
- False Positives: 2
- False Negatives: 0

The Zero False-Negative Guarantee:

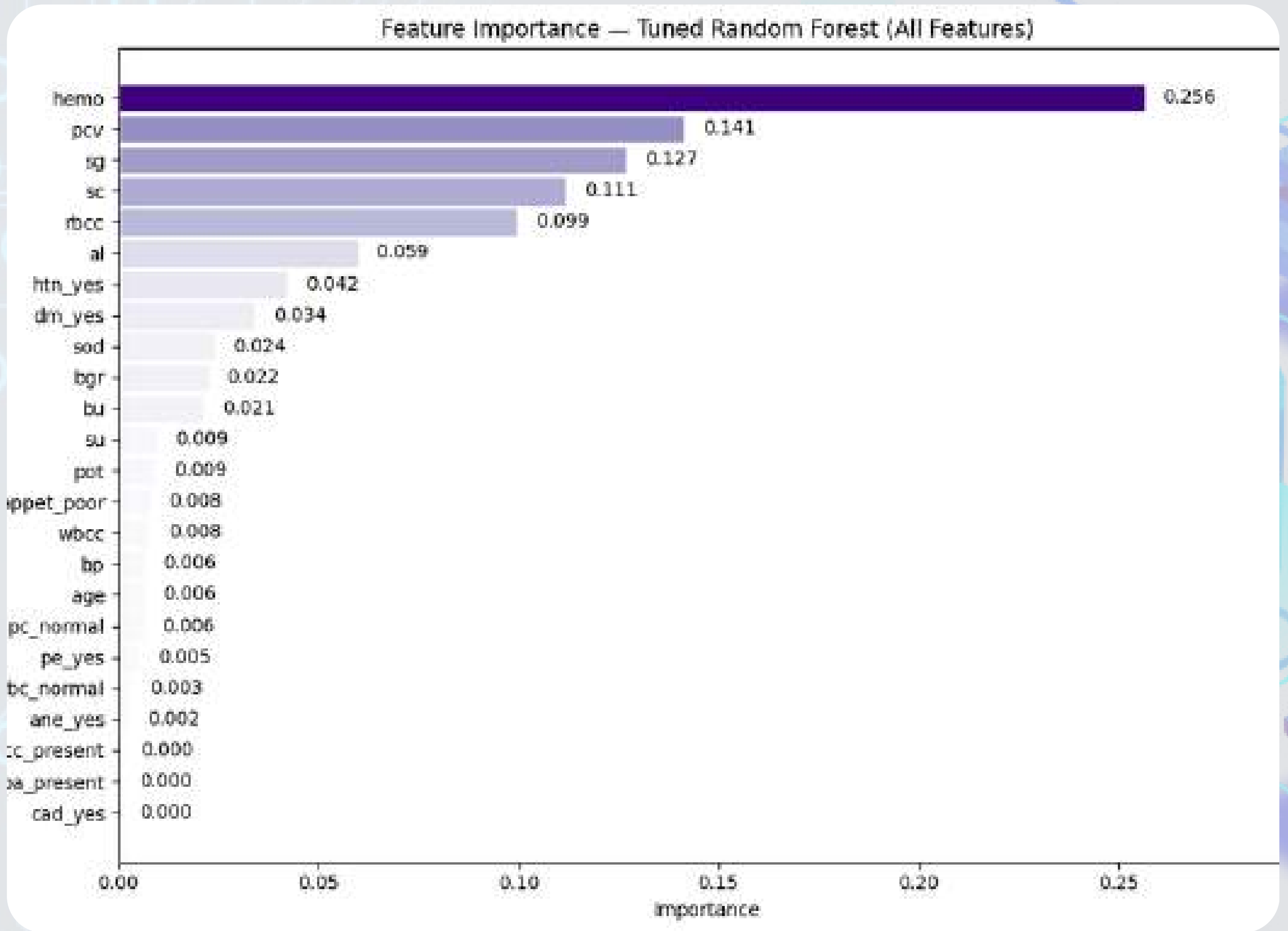
- Sensitivity / Recall on CKD = 100%: In medical screening, missing a positive patient (Type II error) is catastrophic.
- A 6.7% false alarm rate (2 healthy patients flagged for follow-up testing) is an acceptable clinical trade-off to ensure zero undiagnosed patients slip through



Feature Importance

- **Top Predictive Drivers**:
  - Hemoglobin (hemo — 0.256): Strongest single split contributor.
  - Packed Cell Volume (pcv—0.141): Direct proxy for blood concentration and red cell volume.
  - Specific Gravity (sg — 0.127): Urine concentration indicator.
  - Serum Creatinine (sc— 0.111): Primary metric of glomerular clearance rate.

- **Biological Grounding:**
  - The model learned that impaired kidneys fail to produce erythropoietin, triggering renal anemia (hemo, pcv, rbcc account for ~50% total cumulative importance).
  - Structural urine markers (sg, al) combined with waste accumulation (sc) provide orthogonal validation across different stages of kidney damage.



External Validation

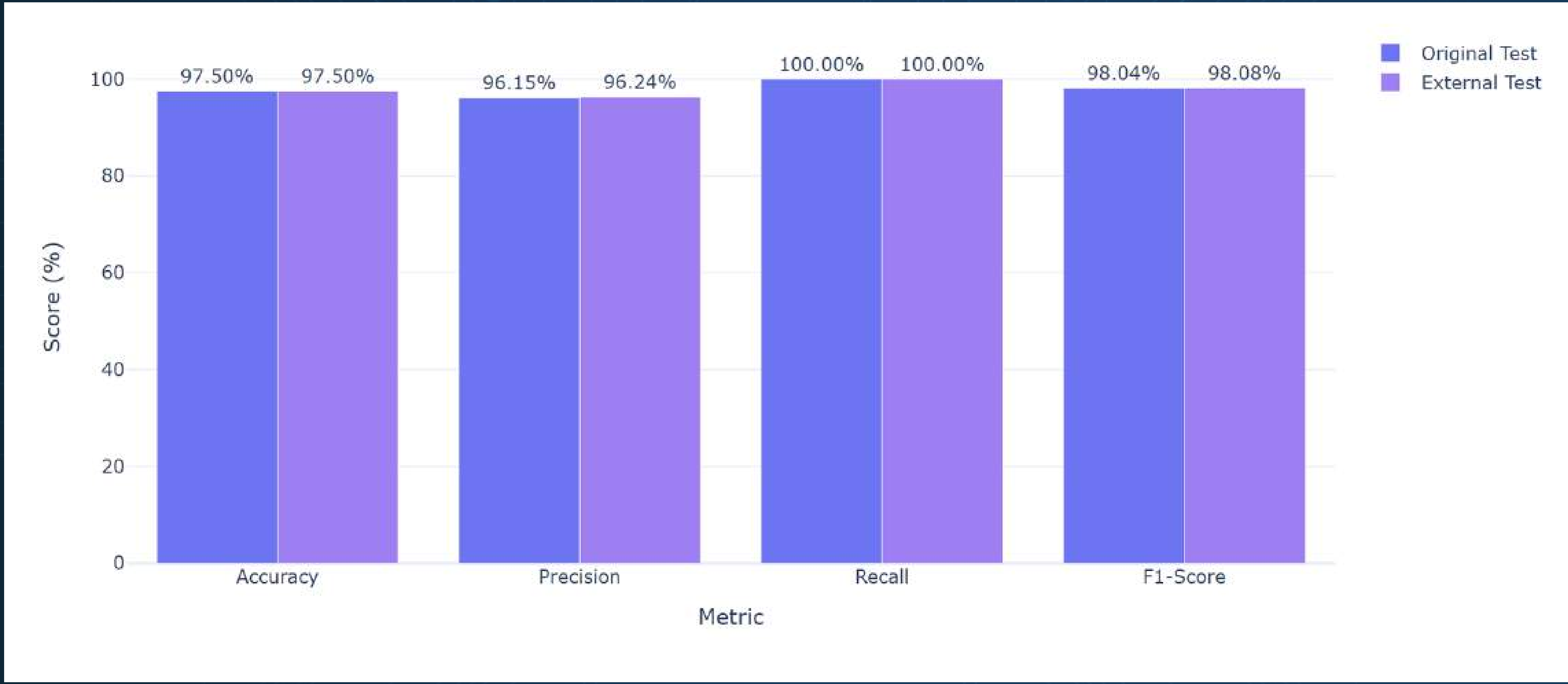
- Independent Test Set: The tuned Random Forest was evaluated on a completely separate 200-patient dataset (128 CKD / 72 No CKD) — never seen during training, tuning, or preprocessing fitting.
-

- Performance Held Up: 97.5% Accuracy, 100% Recall, 96.24% Precision, 98.08% F1 which was nearly identical to the original 80-patient holdout test.

- Zero Missed Diagnoses: All 128 CKD cases were correctly identified; the only errors were 5 healthy patients flagged for unnecessary follow-up.

- Takeaway: Consistent performance across two independent test sets indicates the model is learning real clinical signal (hemoglobin, creatinine, specific gravity) rather than memorizing the original 400-patient dataset.

	Metric	Original Test (%)	External Test (%)	Difference (%)
0	Accuracy	97.50	97.50	0.00
1	Precision	96.15	96.24	0.09
2	Recall	100.00	100.00	0.00
3	F1-Score	98.04	98.08	0.05



Deployment Architecture

Framework & Styling

Built with Streamlit utilizing a custom dark-mode UI layout (stApp), radial gradients, and dynamic badge styling for live/demo states.

Modular Navigation

Sidebar routing enables seamless switching between three distinct clinical evaluation engines within a single unified application.

Production Pipeline

Leverages joblib for dynamic loading of pre-trained model binaries (best_ecg_model.pkl, etc.), scalers, and imputers.

Performance Optimization

Utilizes @st.cache_resource for efficient in-memory model caching and rapid asset retrieval during operations.

ECG Model

Input Flexibility

Supports a "Quick Sample Loader" with predefined arrhythmia presets and a "Batch CSV Upload" for raw 188-sample time-series telemetry.

Interactive Telemetry

Renders live Plotly line charts to visualize waveforms, highlighting R-peak detections and mapping raw codes directly into actionable clinical interpretations.

Heart Failure Survival Risk Evaluator

Multi-Tab Interface

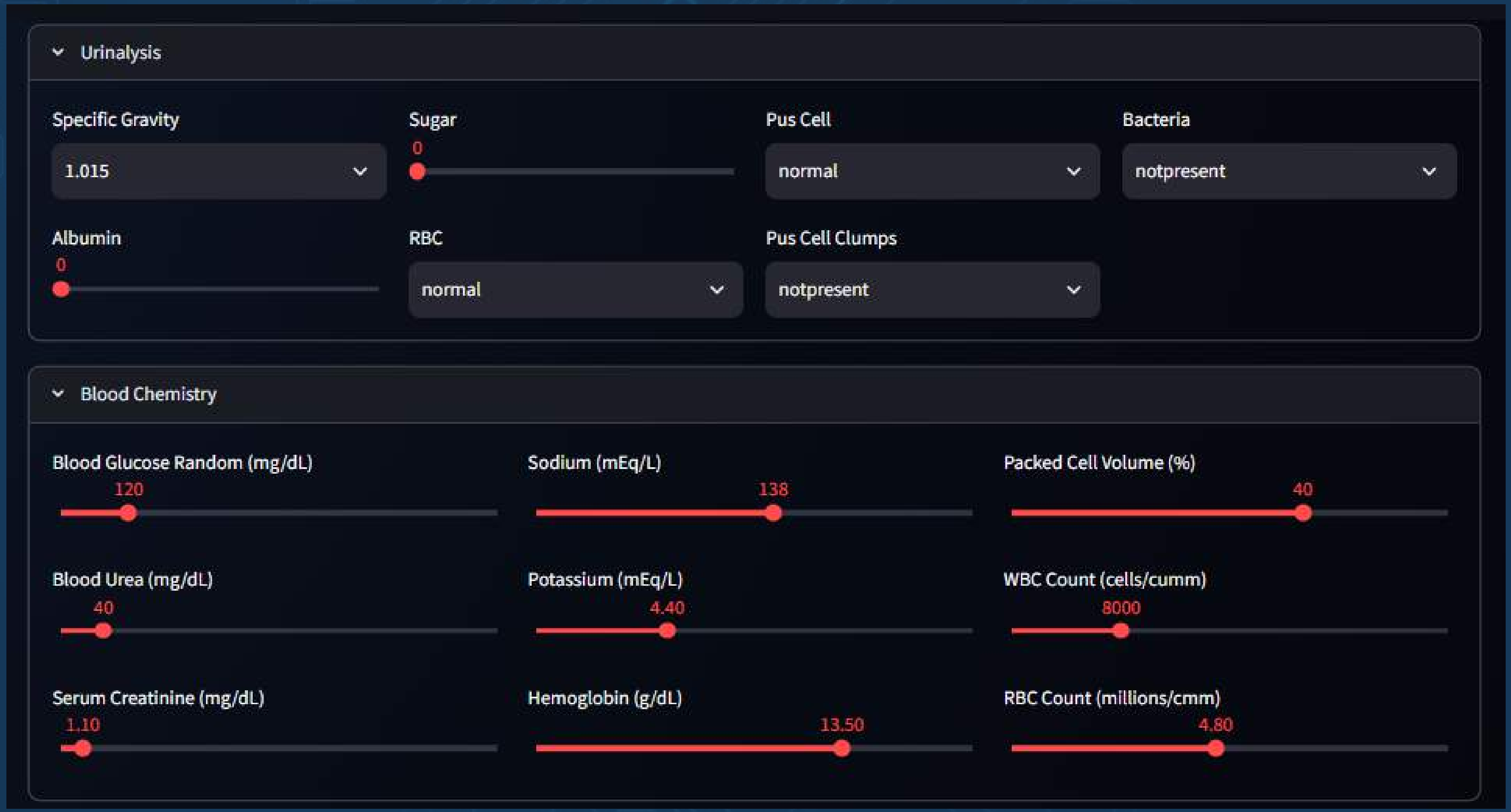
Combines "Manual Patient Assessment" (sliders for ejection fraction, creatinine; toggles for comorbidities) with batch CSV ingestion capabilities.

Risk Stratification

Projects continuous mortality probabilities onto a color-coded Plotly Gauge, categorizing patients into Stable, Moderate Caution, or Critical Risk utilizing selective scaling.

CKD Multi-Parameter Diagnostic

- **Comprehensive Panels Organizes 24 distinct clinical attributes into logical, collapsible sections: Patient Vitals, Urinalysis, Blood Chemistry, and Clinical History.**
- **Defensive Data Wrangling Executes robust background preprocessing, handling missing numeric imputation, categorical one-hot encoding, and strict column alignment.**
- **Diagnostic Output Delivers an immediate clinical verdict (CKD Present vs. Non-CKD / Normal), supported by probability gauges and automated nephrology referral notes.**



Conclusion and Future Challenges

Core Achievements

Unified Diagnostic Suite: Successfully consolidated complex time-series telemetry and tabular clinical data into a single, low-latency web interface.

Zero-Compromise Sensitivity: Prioritized 100% recall in the CKD classifier to eliminate false negatives, ensuring no critical diagnoses are missed during initial screening.

Future Expansion

Mobile Deployment: Migrating to native mobile apps via Flutter for rapid point-of-care access on physician ward rounds.

Edge Processing: Transitioning real-time ECG waveform classification directly to local microprocessors to reduce server latency.

EHR Integration & Drift: Establishing automated telemetry monitoring and secure ingestion pipelines for live clinical data standards.

Thank you